

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-9682
Project Name	AI StudyBuddy – AI-Augmented Backend Development
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

AI StudyBuddy – Technical Architecture (three-tier: React client, Express API, MongoDB)

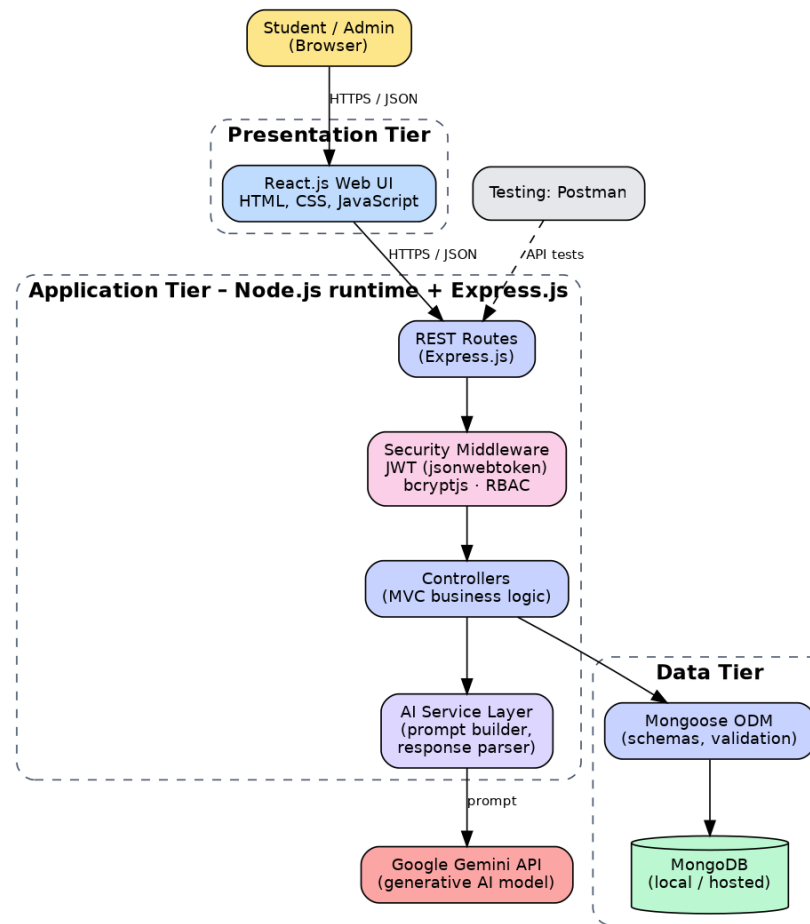


Figure 1: Technical architecture and technology stack of AI StudyBuddy

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1. 1	User Interface	React.js dashboard where students and admins register, log in, upload study material and view AI-generated resources	HTML, CSS, JavaScript, React.js
2. 2	Application Logic-1	REST API server: routing, request handling and MVC controllers	Node.js (v16+), Express.js
3. 3	Application Logic-2	Authentication and authorisation: password hashing, JWT verification and role-based access control	JWT (jsonwebtoken), bcryptjs, RBAC middleware
4. 4	Application Logic-3	AI service layer that builds prompts, calls the AI model and parses summaries, flashcards, quizzes and study plans	Node.js service modules, Google Gemini API SDK / REST
5. 5	Database	NoSQL document store for users, study materials, summaries, flashcards, quizzes and study plans	MongoDB with Mongoose ODM
6. 6	Cloud Database	Hosted MongoDB service for the deployed application (planned)	MongoDB Atlas (or equivalent hosted MongoDB)
7. 7	File Storage	Study material is stored as text content in MongoDB, so no separate file store is required	MongoDB (local filesystem only if file uploads are added)
8. 8	External API-1	Google Gemini API generates summaries, flashcards, quizzes and study plans from the student's content	Google Gemini API
9. 9	External API-2	Not used in the current version	N/A
10. 1 0	Machine Learning Model	Pre-trained generative language model used through the API for natural language tasks; no custom model is trained	Google Gemini (large language model)
11. 1 1	Infrastructure (Server / Cloud)	Local server configuration: Windows 10/11, macOS or Linux, Intel Core i5 / Ryzen 5 or better, 8 GB RAM (16 GB recommended), 1 GB free disk, Node.js v16+, npm v8+, MongoDB, VS Code, Postman. Cloud server configuration: Node.js hosting service plus hosted MongoDB (planned).	Local development server, cloud hosting for deployment

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and libraries used	Node.js, Express.js, React.js, Mongoose, jsonwebtoken, bcryptjs
2.	Security Implementations	JWT-based authentication, bcrypt password hashing, role-based access control, input validation, Gemini API key stored in environment variables, HTTPS recommended in deployment, OWASP guidelines	JWT, bcryptjs, RBAC, dotenv, OWASP Top Ten
3.	Scalable Architecture	Three-tier design (React client, Express API, MongoDB) with a modular MVC structure and a separate AI service layer. Stateless JWT authentication allows horizontal scaling.	3-tier architecture, MVC, REST API
4.	Availability	Stateless API instances can run behind a load balancer, and MongoDB replication is planned for the hosted database. Non-AI features continue if the Gemini API is unavailable.	Load balancer, MongoDB replica set (planned)
5.	Performance	Non-blocking I/O in Node.js, saved AI results reused to avoid repeat Gemini calls, database indexes on userId and materialId, and rate limiting on AI routes.	Node.js async I/O, MongoDB indexes, rate limiting

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

Node.js Documentation: <https://nodejs.org/en/docs>

Express.js Routing Guide: <https://expressjs.com/en/guide/routing.html>

MongoDB Documentation: <https://www.mongodb.com/docs/>

Mongoose ODM Documentation: <https://mongoosejs.com/docs/>

JSON Web Token Introduction: <https://jwt.io/introduction>

Google Gemini API Documentation: <https://ai.google.dev/gemini-api/docs>

React Documentation: <https://react.dev/>